BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Mathematics (Science) (Lahore, Tenth 2021) (Paper - II) (Objective Type) (Group - I) (Session 2017-2019 to 2019-2021)

Time Allowed: 20 Min | **Maximum Marks:** 15

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Two square roots of unity are :

(A) 1, -1
(B) 1, ω
(C) 1, -ω
(D) ω, ω^2

Q2. A circle has only one -------- :

(A) Secant
(B) Chord
(C) Diameter
(D) Centre

Q3. The spread or scatterness of observations in a data set is called : (printed "scatterness" in the source scan)

(A) Average
(B) Central tendency
(C) Dispersion
(D) Mean

Q4. The portion of a circle between two radii and an arc is called :

(A) Chord
(B) Segment
(C) Sector
(D) Diameter

Q5. Two linear factors of $x^2 - 15x + 56$ are :

(A) (x - 7) & (x + 8)
(B) (x + 7) & (x - 8)
(C) (x - 7) & (x - 8)
(D) (x + 7) & (x + 8)

Q6. If $A \subseteq B$, then $A \cup B$ is equal to :

(A) A
(B) B
(C) φ
(D) A - B

Q7. Line segment joining any point of the circle to the centre is called :

(A) Radial segment
(B) Diameter
(C) Circumference
(D) Perimeter

Q8. In a proportion a : b :: c : d , b and c are called:

(A) Means
(B) Extremes
(C) Third proportional
(D) Fourth proportional

Q9. The arcs opposite to incongruent central angles of a circle are always :

(A) Congruent
(B) Incongruent
(C) Parallel
(D) Perpendicular

Q10. The domain of R. = {(0, 2), (2, 3), (3, 3), (3, 4)} is :

(A) {0, 3, 4}
(B) {2, 3, 4}
(C) {0, 2, 4}
(D) {0, 2, 3}

Q11. If α, β are the roots of $x^2 - x - 1 = 0$, then product of the roots 2α and 2β is :

(A) -2
(B) 2
(C) 4
(D) -4

Q12. If $y^2 \propto 1/x^3$, then :

(A) $y^2 = k/x^3$
(B) $y^2 = 1/x^3$
(C) $y^2 = x^3$
(D) $y^2 = kx^3$

Q13. A histogram is a set of adjacent :

(A) Squares
(B) Rectangles
(C) Circles
(D) Triangles

Q14. Product of cube roots of unity is :

(A) 0
(B) 1
(C) -1
(D) 3

Q15. 3π/4 radians = :

(A) 115°

(B) 30°

(C) 150°

(D) 135°

Mathematics (Science) (Lahore, Tenth 2021) (Paper - II) (Subjective Type) (Group - I) (Session 2017-2019 to 2019-2021)

Time Allowed: 2.10 hrs | **Maximum Marks:** 60

QPART - I.

Q2. (Marks 12) Write short answers to any SIX (6) questions:

i Define reciprocal equation.

ii Solve by factorization: $x^2 - 11x = 152$

iii Solve: $x^2 + 2x - 2 = 0$

iv Evaluate $(1 - 3\omega - 3\omega^2)^5$

v Find the product of complex cube roots of unity.

vi If the ratios $3x + 1 : 6 + 4x$ and $2 : 5$ are equal, find the value of x.

vii If $y \propto 1/x$ and $y = 4$, when $x = 3$, find x when $y = 24$.

viii Find ω^2, if $\omega = (-1 + \sqrt{-3})/2$

ix Find a third proportional to $(x - y)^2$, $x^2 - y^2$

Q3. (Marks 12) Write short answers to any SIX (6) questions:

i Resolve into partial fractions: $(x - 11)/((x - 4)(x + 3))$

ii What are partial fractions?

iii If $X = \{1, 4, 7, 9\}$ and $Y = \{2, 4, 5, 9\}$, then find $Y \cap X$

iv Define an Onto function.

v If A = N and B = W then find the value of B - A.

vi If $L = \{a, b, c\}$, $M = \{d, e, f, g\}$, then find two binary relation in $L \times M$.

vii Find the arithmetic mean by direct method for the set of data : 200,225, 350,375, 270,320,290

viii Define class mark.

ix Name two measures of central tendency.

Q4. (Marks 12) Write short answers to any SIX (6) questions:

i Find 'r', when l = 56 cm and θ = 45°

ii Define radian measure of an angle.

iii Express the angle 315° into radian.

iv State theorem of componendo and dividendo.

v Find the fourth proportional to 8, 7, 6.

vi In a ΔABC, a = 17 cm, b = 15 cm and c = 8 cm, find m ∠B.

vii Divide an arc of any length into four equal parts.

viii Write the closes quadrantal angles between which the angle -3 π/4 lies. (printed "closes" in the source scan)

ix Verify: $(1-\sin\theta)(1+\sin\theta) = \cos^2\theta$

QPART - II. Note : Attempt THREE questions in all. But question No.9 is Compulsory.

Q5.a. (Marks 4) Solve by factorization: $(x + 1)/x + x/(x + 1) = 25/12$

Q5.b. (Marks 4) Find m, if the roots of the equation $x^2 + 7x + 3m - 5 = 0$ satisfy the relation $3\alpha - 2\beta = 4$

Q6.a. (Marks 4) If a : b = c : d (a, b, c, d ≠ 0) then show that $a/b = \sqrt{(a^2 + c^2)}/\sqrt{(b^2 + d^2)}$

Q6.b. (Marks 4) Resolve into partial fractions: $(3x - 11)/((x + 3)(x^2 + 1))$

Q7.a. (Marks 4) If U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, A = {1, 3, 5, 7, 9} and B = {2, 3, 5, 7} then verify $(A \cup B)' = A' \cap B'$

Q7.b. (Marks 4) Find the standard deviation "S" = : 9, 3, 8, 8, 9, 8, 9, 18

Q8.a. (Marks 4) Verify the identity: $(1 + \sin\theta)/(1 - \sin\theta) - (1 - \sin\theta)/(1 + \sin\theta) = 4 \tan\theta \sec\theta$

Q8.b. (Marks 4) Draw circle which touches both the arms of angle : 45°

Q9. (Marks 8) A straight line, drawn from the centre of a circle to bisect a chord (which is not a diameter) is perpendicular to the chord. OR If the angles subtended by two chords of a circle (or congruent circles) at the centre (corresponding centres) are equal, the chords are equal.

GROUP – II

Mathematics (Science) (Lahore, Tenth 2021) (Paper - II) (Objective Type) (Group - II) (Session 2017-2019 to 2019-2021)

Time Allowed: 20 Min | **Maximum Marks:** 15

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. $\text{cosec}^2\theta - \cot^2\theta =$ -------- :

(A) $\tan\theta$ (B) 0

(C) -1 (D) 1

Q2. If $y^2 \propto 1/x^3$, then :

(A) $y^2 = k/x^3$ (B) $y^2 = 1/x^3$

(C) $y^2 = x^2$ (D) $y^2 = kx^3$

Q3. A histogram is a set of adjacent :

(A) Squares (B) Rectangles

(C) Circles (D) Data

Q4. The length of the diameter of a circle is how many times the radius of the circle :

(A) 4 times (B) 3 times

(C) 1 time (D) 2 times

Q5. The number of elements in power set {1,2,3} is :

(A) 4 (B) 6

(C) 8 (D) 9

Q6. Two square roots of unity are :

(A) 1, -1 (B) 1, ω

(C) 1, $-\omega$ (D) ω, ω^2

Q7. A line which has two points in common with a circle is called :

(A) Sine of a circle (B) Cosine of a circle

(C) Tangent of a circle (D) Secant of a circle

Q8. The number of methods to solve a quadratic equation is :

(A) 1 (B) 2

(C) 3 (D) 4

Q9. $(x^3 + 1)/((x - 1)(x + 2))$ is ---------:

(A) A proper fraction
(B) An improper fraction
(C) An identity
(D) A constant term

Q10. Cube roots of -1 are:

(A) $-1, \omega, \omega^2$
(B) $-1, \omega, -\omega^2$
(C) $1, \omega, \omega^2$
(D) $1, -\omega, -\omega^2$

Q11. If $A \subseteq B$, then $A \cap B$ is equal to :

(A) A
(B) B
(C) φ
(D) $\{\varphi\}$

Q12. The semi circumference and the diameter of a circle both subtend a central angle of :

(A) 90°
(B) 180°
(C) 270°
(D) 360°

Q13. The most frequent occurring observation in a data set is called :

(A) Mode
(B) Median
(C) Harmonic mean
(D) Mean

Q14. Find x in proportion 4 : x :: 5 : 15 :

(A) 75/4
(B) 4/3
(C) 3/4
(D) 12

Q15. Through how many non-collinear points can a circle pass:

(A) One
(B) Two
(C) Three
(D) Four

Mathematics (Science) (Lahore, Tenth 2021) (Paper - II) (Subjective Type) (Group - II) (Session 2017-2019 to 2019-2021)

Time Allowed: 2.10 hrs | **Maximum Marks:** 60

QPART - I.

Q2. (Marks 12) Write short answers to any SIX (6) questions:

i Define reciprocal equation.

ii Solve: $(2x - 1/2)^2 = 9/4$

iii Solve: $\sqrt{3}x^2 + x - 4\sqrt{3} = 0$

iv Discuss the nature of roots of equation $3x^2 + 7x - 13 = 0$

v Find w^2, if $w = (-1 + \sqrt{-3})/2$

vi Write the quadratic equation having roots -1, -7

vii If the ratios $3x + 1 : 6 + 4x$ and $2 : 5$ are equal, find the value of x.

viii $a \propto c/b^2$ and $a = 3$ when $b = 4$, find a when $b = 8$

ix Find a mean proportional between $x^2 - y^2$, $(x - y)/(x + y)$

Q3. (Marks 12) Write short answers to any SIX (6) questions:

i Define a rational fraction.

ii Resolve into partial fractions : $(x - 11)/((x - 4)(x + 3))$

iii State De Morgan's laws.

iv If $X = \{1,4,7,9\}$, $Y = \{2,4,5,9\}$, then find $X \cap Y$

v If A = N, B = W, then find B - A.

vi Find a and b if $(3 - 2a, b - 1) = (a - 7, 2b + 5)$

vii Define mode.

viii What is a histogram?

ix Define standard deviation.

Q4. (Marks 12) Write short answers to any SIX (6) questions:

i Define ratio and give one example.

ii Find the third proportional to 28 and 4.

iii Express 315.18° into D°, M' and S" form.

iv Convert $7\pi/8$ into degree.

v Prove that $(1 - \sin^2\theta)(1 + \tan^2\theta) = 1$

vi Find θ when $l = 4.5$ m, $r = 2.5$ m

vii Express the angle into radian : 135°

viii In a ΔABC , $a = 17$ cm, $b = 15$ cm and $c = 8$ cm, find m $\angle B$

ix Divide an arc of any length into four equal parts.

QPART - II. Note: Attempt THREE questions in all. But question no. 9 is compulsory.

Q5.(a). (Marks 4) Solve the equation by completing square: $7x^2 + 2x - 1 = 0$

Q5.(b). (Marks 4) Solve by using synthetic division, if 3 and -4 are the roots of the equation $x^4 + 2x^3 - 13x^2 - 14x + 24 = 0$

Q6.(a). (Marks 4) Using theorem of componendo-dividendo find the value of: $(x - 3y)/(x + 3y) = (x + 3z)/(x - 3z)$ if $x = 3yz/(y - z)$

Q6.(b). (Marks 4) Resolve into partial fractions : $(x^2 + 7x + 11)/((x + 2)^2(x + 3))$

Q7.(a). (Marks 4) If $U = \{1,2,3,4,5,6,7,8,9,10\}$, $A = \{1,3,5,7,9\}$, $B = \{1,4,7,10\}$ then prove that $(A - B)' = A' \cup B$.

Q7.(b). (Marks 4) Find the standard deviation of five teachers' salaries in rupees : 11500, 12400, 15000, 14500, 14800

Q8.(a). (Marks 4) Prove that : $(1 + \sin\theta)/(1 - \sin\theta) - (1 - \sin\theta)/(1 + \sin\theta) = 4 \tan\theta \sec\theta$

Q8.(b). (Marks 4) Draw two equal circles of each radius 2.4 cm. If the distance between their centres is 6 cm, then draw their transverse tangents.

Q9. (Marks 8) A straight line drawn from the centre of a circle to bisect a chord (which is not a diameter) is perpendicular to the chord. OR Two tangents drawn to a circle from a point outside it, are equal in length.